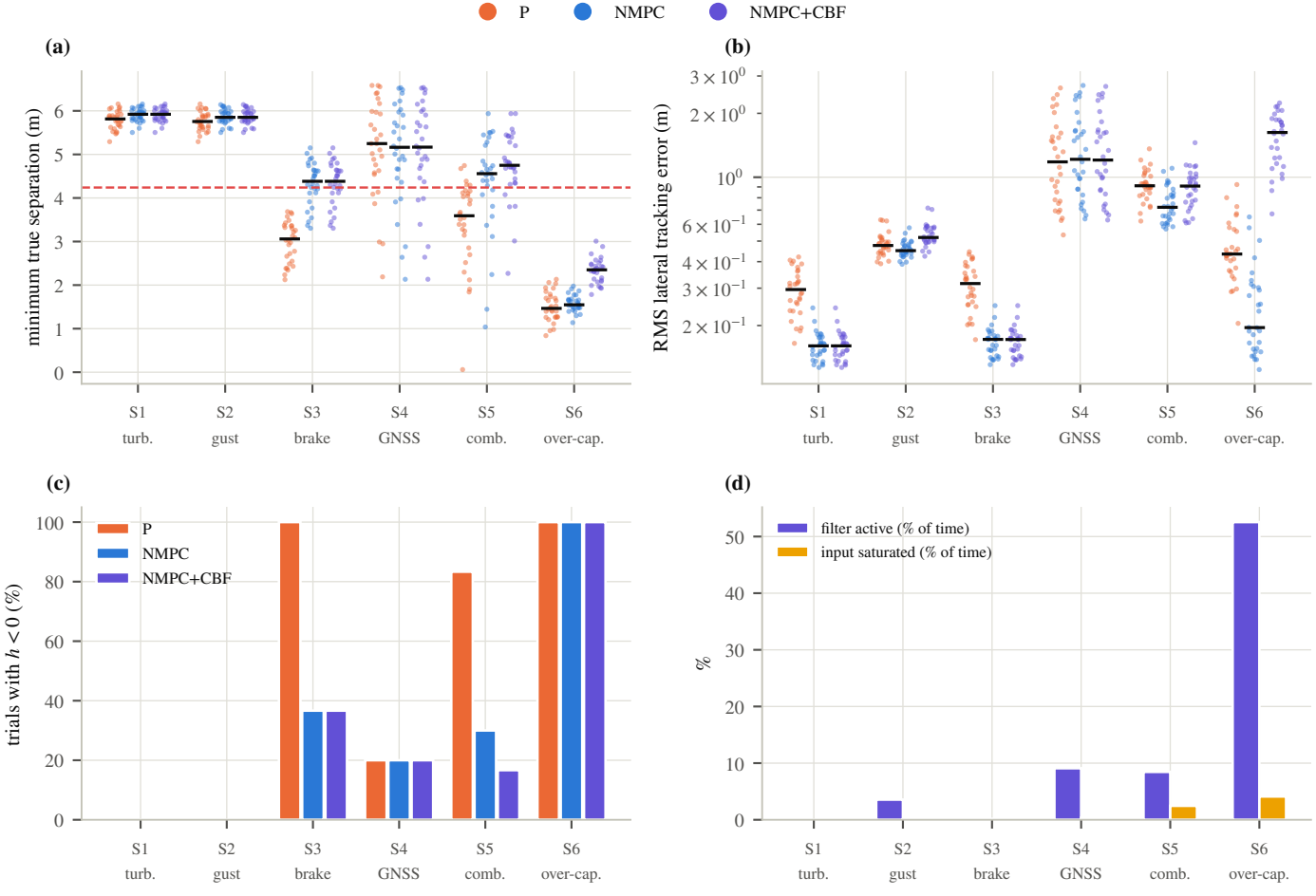
P
NMPC
NMPC+CBF
(a)
minimum true separation (m)
0
1
2
3
4
5
6
S1
turb.
S2
gust
S3
brake
S4
GNSS
S5
comb.
S6
over-cap.
(b)
RMS lateral tracking error (m)
3×10^0
2×10^0
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
S1
turb.
S2
gust
S3
brake
S4
GNSS
S5
comb.
S6
over-cap.
(c)
trials with $h < 0$ (%)
P
NMPC
NMPC+CBF
0
20
40
60
80
100
S1
turb.
S2
gust
S3
brake
S4
GNSS
S5
comb.
S6
over-cap.
(d)
%
filter active (% of time)
input saturated (% of time)
0
10
20
30
40
50
S1
turb.
S2
gust
S3
brake
S4
GNSS
S5
comb.
S6
over-cap.